THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 154 cm²	$\frac{22}{7} \times 7 \times 7$; Area $= 154$ cm^2
2	**(b)** 44 cm	$2 \times \frac{22}{7} \times 7$; Circumference $= 44$ cm
3	**(d)** 7 cm	$r = \frac{44 \times 7}{2 \times 22}$; $r = 7$ cm
4	**(b)** 7 cm	$r^2 = \frac{154 \times 7}{22} = 49$; $r = 7$ cm
5	**(d)** 231 cm²	$\frac{60}{360} \times \frac{22}{7} \times 21^2$; Area $= 231$ cm^2
6	**(b)** 22 cm	$\frac{60}{360} \times 2 \times \frac{22}{7} \times 21$; Arc $= 22$ cm
7	**(d)** 36 cm	Arc $= \pi r = 22$ cm; diameter $= 14$ cm; Perimeter $= 36$ cm
8	**(d)** 25 cm	Quarter arc $= \frac{\pi r}{2} = 11$ cm; $2r = 14$ cm; Perimeter $= 25$ cm
9	**(a)** 616 cm²	$\frac{22}{7} \times 14 \times 14$; Area $= 616$ cm^2
10	**(b)** 88 cm	$2 \times \frac{22}{7} \times 14$; Circumference $= 88$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area of a circle $= \pi r^2$; $\frac{22}{7} \times 3.5 \times 3.5$ (1); Area $= 38.5$ m^2 (1).	1+1
12	Circumference $= 2\pi r$; $2 \times \frac{22}{7} \times 3.5$ (1); Circumference $= 22$ m (1).	1+1
13	Area of sector $= \frac{\theta}{360} \times \pi r^2$; $\frac{45}{360} \times \frac{22}{7} \times 21^2$ (1); Area $= 173.25$ cm^2 (1).	1+1
14	Area of a circle $= \pi r^2$; $\frac{22}{7} \times 10.5 \times 10.5$ (1); Area $= 346.5$ m^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Arc length $= \frac{\theta}{360} \times 2\pi r$ (1); $\frac{45}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc $= 16.5$ cm (1).	1+1+1
16	Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{60}{360} \times 2\pi \times 21 = 22$ cm (1); Perimeter $= 22 + 42 = 64$ cm (1).	1+1+1
17	Area of semicircle $= \frac{1}{2}\pi r^2$ (1); $\frac{1}{2} \times \frac{22}{7} \times 7^2$ (1); Area $= 77$ cm^2 (1).	1+1+1
	OR: Arc length $= \frac{\theta}{360} \times 2\pi r$ (1); $\frac{72}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc $= 26.4$ cm (1).	1+1+1
18	Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{90}{360} \times 2\pi \times 21 = 33$ cm (1); Perimeter $= 33 + 42 = 75$ cm (1).	1+1+1
	OR: Area of semicircle $= \frac{1}{2}\pi r^2$ (1); $\frac{1}{2} \times \frac{22}{7} \times 14^2$ (1); Area $= 308$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Outer radius $= 21$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(21^2 - 14^2)$ (2); Area of path $= 770$ m^2 (1).	2+2+1
	OR: The largest circle has diameter equal to the side, so $r = \frac{s}{2}$ (2); Circle area $= \frac{22}{7} \times 7^2 = 154$ cm^2; square $= 196$ cm^2 (2); Left over $= 42$ cm^2 (1).	2+2+1
20	Outer radius $= 28$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(28^2 - 21^2)$ (2); Area of path $= 1078$ m^2 (1).	2+2+1
	OR: The largest circle has diameter equal to the side, so $r = \frac{s}{2}$ (2); Circle area $= \frac{22}{7} \times 14^2 = 616$ cm^2; square $= 784$ cm^2 (2); Left over $= 168$ cm^2 (1).	2+2+1

Mistakes to avoid:

- Using πr^2 in the arc-length formula or $2\pi r$ in the sector-area formula.
- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.
- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 14 cm	$r = \frac{88\times7}{2\times22}$; $r = 14$ cm
2	**(c)** 14 cm	$r^2 = \frac{616\times7}{22} = 196$; $r = 14$ cm
3	**(c)** 346.5 cm²	$\frac{90}{360} \times \frac{22}{7} \times 21^2$; Area $= 346.5$ cm^2
4	**(d)** 33 cm	$\frac{90}{360} \times 2 \times \frac{22}{7} \times 21$; Arc $= 33$ cm
5	**(d)** 72 cm	Arc $= \pi r = 44$ cm; diameter $= 28$ cm; Perimeter $= 72$ cm
6	**(b)** 50 cm	Quarter arc $= \frac{\pi r}{2} = 22$ cm; $2r = 28$ cm; Perimeter $= 50$ cm
7	**(a)** 1386 cm²	$\frac{22}{7} \times 21 \times 21$; Area $= 1386$ cm^2
8	**(b)** 132 cm	$2 \times \frac{22}{7} \times 21$; Circumference $= 132$ cm
9	**(a)** 21 cm	$r = \frac{132\times7}{2\times22}$; $r = 21$ cm
10	**(a)** 21 cm	$r^2 = \frac{1386\times7}{22} = 441$; $r = 21$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Circumference $= 2\pi r$; $2 \times \frac{22}{7} \times 10.5$ (1); Circumference $= 66$ m (1).	1+1
12	Area of sector $= \frac{\theta}{360} \times \pi r^2$; $\frac{72}{360} \times \frac{22}{7} \times 21^2$ (1); Area $= 277.2$ cm^2 (1).	1+1
13	Area of a circle $= \pi r^2$; $\frac{22}{7} \times 17.5 \times 17.5$ (1); Area $= 962.5$ m^2 (1).	1+1
14	Circumference $= 2\pi r$; $2 \times \frac{22}{7} \times 17.5$ (1); Circumference $= 110$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Arc length $= \frac{\theta}{360} \times 2\pi r$ (1); $\frac{150}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc $= 55$ cm (1).	1+1+1
16	Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{120}{360} \times 2\pi \times 21 = 44$ cm (1); Perimeter $= 44 + 42 = 86$ cm (1).	1+1+1
17	Area of semicircle $= \frac{1}{2}\pi r^2$ (1); $\frac{1}{2} \times \frac{22}{7} \times 21^2$ (1); Area $= 693$ cm^2 (1).	1+1+1
	OR: Arc length $= \frac{\theta}{360} \times 2\pi r$ (1); $\frac{108}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc $= 39.6$ cm (1).	1+1+1
18	Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{30}{360} \times 2\pi \times 21 = 11$ cm (1); Perimeter $= 11 + 42 = 53$ cm (1).	1+1+1
	OR: Area of semicircle $= \frac{1}{2}\pi r^2$ (1); $\frac{1}{2} \times \frac{22}{7} \times 28^2$ (1); Area $= 1232$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Outer radius $= 35$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(35^2 - 28^2)$ (2); Area of path $= 1386$ m^2 (1).	2+2+1
	OR: The largest circle has diameter equal to the side, so $r = \frac{s}{2}$ (2); Circle area $= \frac{22}{7} \times 21^2 = 1386$ cm^2; square $= 1764$ cm^2 (2); Left over $= 378$ cm^2 (1).	2+2+1
20	Outer radius $= 42$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(42^2 - 35^2)$ (2); Area of path $= 1694$ m^2 (1).	2+2+1
	OR: The largest circle has diameter equal to the side, so $r = \frac{s}{2}$ (2); Circle area $= \frac{22}{7} \times 10.5^2 = 346.5$ cm^2; square $= 441$ cm^2 (2); Left over $= 94.5$ cm^2 (1).	2+2+1

Mistakes to avoid:

- Using πr^2 in the arc-length formula or $2\pi r$ in the sector-area formula.
- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.
- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 462 cm²	$\frac{120}{360} \times \frac{22}{7} \times 21^2$; Area $= 462$ cm^2
2	**(c)** 44 cm	$\frac{120}{360} \times 2 \times \frac{22}{7} \times 21$; Arc $= 44$ cm
3	**(c)** 108 cm	Arc $= \pi r = 66$ cm; diameter $= 42$ cm; Perimeter $= 108$ cm
4	**(b)** 75 cm	Quarter arc $= \frac{\pi r}{2} = 33$ cm; $2r = 42$ cm; Perimeter $= 75$ cm
5	**(a)** 2464 cm²	$\frac{22}{7} \times 28 \times 28$; Area $= 2464$ cm^2
6	**(b)** 220 cm	$2 \times \frac{22}{7} \times 35$; Circumference $= 220$ cm
7	**(c)** 28 cm	$r = \frac{176 \times 7}{2 \times 22}$; $r = 28$ cm
8	**(d)** 28 cm	$r^2 = \frac{2464 \times 7}{22} = 784$; $r = 28$ cm
9	**(b)** 693 cm²	$\frac{180}{360} \times \frac{22}{7} \times 21^2$; Area $= 693$ cm^2
10	**(b)** 66 cm	$\frac{180}{360} \times 2 \times \frac{22}{7} \times 21$; Arc $= 66$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area of sector $= \frac{\theta}{360} \times \pi r^2$; $\frac{150}{360} \times \frac{22}{7} \times 21^2$ (1); Area $= 577.5$ cm^2 (1).	1+1
12	Area of a circle $= \pi r^2$; $\frac{22}{7} \times 24.5 \times 24.5$ (1); Area $= 1886.5$ m^2 (1).	1+1
13	Circumference $= 2\pi r$; $2 \times \frac{22}{7} \times 24.5$ (1); Circumference $= 154$ m (1).	1+1
14	Area of sector $= \frac{\theta}{360} \times \pi r^2$; $\frac{40}{360} \times \frac{22}{7} \times 21^2$ (1); Area $= 154$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Arc length $= \frac{\theta}{360} \times 2\pi r$ (1); $\frac{36}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc $= 13.2$ cm (1).	1+1+1
16	Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{180}{360} \times 2\pi \times 21 = 66$ cm (1); Perimeter $= 66 + 42 = 108$ cm (1).	1+1+1
17	Area of semicircle $= \frac{1}{2}\pi r^2$ (1); $\frac{1}{2} \times \frac{22}{7} \times 35^2$ (1); Area $= 1925$ cm^2 (1).	1+1+1
	OR: Arc length $= \frac{\theta}{360} \times 2\pi r$ (1); $\frac{24}{360} \times 2 \times \frac{22}{7} \times 21$ (1); Arc $= 8.8$ cm (1).	1+1+1
18	Perimeter $=$ arc $+ 2r$ (1); Arc $= \frac{150}{360} \times 2\pi \times 21 = 55$ cm (1); Perimeter $= 55 + 42 = 97$ cm (1).	1+1+1
	OR: Area of semicircle $= \frac{1}{2}\pi r^2$ (1); $\frac{1}{2} \times \frac{22}{7} \times 42^2$ (1); Area $= 2772$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Outer radius $= 28$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(28^2 - 14^2)$ (2); Area of path $= 1848$ m^2 (1).	2+2+1
	OR: The largest circle has diameter equal to the side, so $r = \frac{s}{2}$ (2); Circle area $= \frac{22}{7} \times 17.5^2 = 962.5$ cm^2; square $= 1225$ cm^2 (2); Left over $= 262.5$ cm^2 (1).	2+2+1
20	Outer radius $= 14$ m (2); Area $= \pi(R^2 - r^2) = \frac{22}{7}(14^2 - 7^2)$ (2); Area of path $= 462$ m^2 (1).	2+2+1
	OR: The largest circle has diameter equal to the side, so $r = \frac{s}{2}$ (2); Circle area $= \frac{22}{7} \times 3.5^2 = 38.5$ cm^2; square $= 49$ cm^2 (2); Left over $= 10.5$ cm^2 (1).	2+2+1

Mistakes to avoid:

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